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The sin-omial integral for $\binom{r}{s}$: the area of a level set is a half-residue

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Theorem 1. For real $r > 0$ and $0 \le s \le r$,

$$\frac{1}{\pi}\int_0^\pi \frac{\sin^r x}{\sin^s\!\left(\frac{sx}{r}\right)\,\sin^{r-s}\!\left(\frac{(r-s)x}{r}\right)}\,dx = \binom{r}{s} := \frac{\Gamma(r+1)}{\Gamma(s+1)\,\Gamma(r-s+1)}.$$

If $s \in \{0, r\}$ the integrand is $\equiv 1$ ($\sin^0 := 1$) and both sides are 1. Otherwise let $a = s/r \in (0,1)$, $b = 1-a$, $K(x) = \frac{\sin x}{\sin^a(ax)\sin^b(bx)}$ (so the integrand is K^r), $\rho(x) = \frac{\sin ax}{\sin bx} = \frac{a\,S(ax)}{b\,S(bx)}$ with $S(u) = \frac{\sin u}{u}$ (so ρ is real-analytic and positive on $[0,\pi]$, and $\rho(0) = \frac{a}{b}$), $g(t) = t^a(1-t)^b$, and $\Phi(t,x) = g(t)K(x)$ on $R = (0,1)\times(0,\pi)$.

Theorem 2. $|\{\Phi > \lambda\}| = \pi(1-\lambda)$ for $0 \le \lambda \le 1$; that is, Φ is uniform on $(0,1)$ for the measure $dt\,dx/\pi$ on R. (For $a = \frac12$ this is Archimedes' hat-box theorem.)

Theorem 2 ⇒ Theorem 1. By the layer-cake formula, $\frac{1}{\pi}\iint_R \Phi^r = \int_0^1 r\lambda^{r-1}(1-\lambda)\,d\lambda = \frac{1}{r+1}$. By Fubini and Euler's Beta integral, $\frac{1}{\pi}\iint_R \Phi^r = B(ar+1, br+1)\cdot\frac{1}{\pi}\int_0^\pi K^r$. Hence $\frac{1}{\pi}\int_0^\pi K^r = \frac{\Gamma(r+2)}{(r+1)\Gamma(ar+1)\Gamma(br+1)} = \binom{r}{s}$. □

Idea of the proof of Theorem 2. On the level curve $\Phi = \lambda$, the law of sines makes t a meromorphic function of the coordinate ξ of (b), real on $\mathbb{R}$. Its only pole in the strip $|\operatorname{Im}\xi| < \pi$ is simple, with residue $-(1-\lambda)$, at the image of the maximum point $(a,0)$. The curve joins two real points, halfway around the pole, so the area is $\pi(1-\lambda)$.

Proof of Theorem 2. **(a) Law of sines.** Let $z = \rho e^{ix}$ and $z^a := \rho^a e^{iax}$. The identity $\sin bx + \sin ax\, e^{ix} = \sin x\, e^{iax}$ (its real part is $\sin bx = \sin(x-ax)$) and $K\rho^a = \frac{\sin x}{\sin bx}$ give $1 + z = K(x)\,z^a$ on $[0,\pi]$.

(b) Key identity. For $(t,x) \in (0,1)\times[0,\pi)$ let $\xi = \log\frac{(1-t)\rho(x)}{t} + ix$. Then $e^\xi = \frac{1-t}{t}\,z$ and $e^{a\xi} = (\frac{1-t}{t})^a z^a$, so (a) gives $\Phi e^{a\xi} = (1-t)(1+z) = 1 - t + te^\xi$, that is,

$$t\,(1-e^\xi) = 1 - \Phi(t,x)\,e^{a\xi}. \tag{$*$}$$

On $x = 0$, $\xi = \log\frac{(1-t)a}{tb}$ is real, decreasing in t, and 0 at $t = a$. Elsewhere $e^\xi \ne 1$.

(c) The region $D_\lambda = \{\Phi > \lambda\}$, $0 < \lambda < 1$. g increases on $[0,a]$ and decreases on $[a,1]$. K decreases from $K(0) = a^{-a}b^{-b}$ to $K(\pi) = 0$ (as in Kanter 1975): with $c(y) = y\cot y$, which decreases because $c'(y) = \frac{\sin 2y - 2y}{2\sin^2 y} < 0$, we have $x\,(\log K)' = c(x) - a\,c(ax) - b\,c(bx) < 0$. So $\Phi < g(a)K(0) = 1$ on R, $\nabla\Phi \ne 0$ on $(0,1)\times[0,\pi)$ except at $(a,0)$, and $D_\lambda = \{0 < x < x_\lambda,\ t_-(x) < t < t_+(x)\}$. Here $g(a)K(x_\lambda) = \lambda$, and $t_-(x) \le a \le t_+(x)$ are the roots of $\Phi(\cdot,x) = \lambda$. The level arc $\Gamma_\lambda = \{(t_\pm(x), x) : 0 \le x \le x_\lambda\}$ is C^1; orient it from $(t_+(0),0)$ to $(t_-(0),0)$. By Fubini, $|D_\lambda| = \int_0^{x_\lambda}(t_+ - t_-)\,dx = \int_{\Gamma_\lambda} t\,dx$.

(d) Half-residue. On Γ_λ, $(*)$ gives

$$t = \tau_\lambda(\xi) := \frac{1-\lambda e^{a\xi}}{1-e^\xi} = \frac{1-\lambda}{1-e^\xi} + \lambda\,\psi(\xi), \qquad \psi(\xi) := \frac{1-e^{a\xi}}{1-e^\xi} = \frac{E_a(\xi)}{E_1(\xi)}, \qquad E_c(\xi) := \frac{e^{c\xi}-1}{\xi} = \sum_{k\ge0}\frac{c^{k+1}\xi^k}{(k+1)!}.$$

As t is real, $t\,dx = \operatorname{Im}(t\,d\xi)$, so $|D_\lambda| = \operatorname{Im}\int_{\gamma_\lambda}\tau_\lambda\,d\xi$ with $\gamma_\lambda = \xi(\Gamma_\lambda)$. By (b), γ_λ is a C^1 path from $\xi_+ < 0$ to $\xi_- > 0$ (both real) and otherwise lies in $0 < \operatorname{Im}\xi < \pi$. There $1 - e^\xi$ lies in the lower half-plane, so $\xi - \log(1-e^\xi)$, with $\arg(1-e^\xi) \in (-\pi, 0)$, is a primitive of $\frac{1}{1-e^\xi}$. Its imaginary part $x - \arg(1-e^\xi)$ goes from 0 at ξ_+ (where $1 - e^\xi > 0$) to π at ξ_- (where $1 - e^\xi < 0$). And ψ is holomorphic on the convex strip $|\operatorname{Im}\xi| < \pi$, since E_a, E_1 are entire and $E_1 \ne 0$ there (its zeros are $2\pi ik$, $k \ne 0$); $\psi(0) = a$ and ψ is real on $\mathbb{R}$. So its primitive $H(\xi) = \xi\int_0^1 \psi(s\xi)\,ds$ (indeed $H' = \int_0^1 \partial_s(s\,\psi(s\xi))\,ds = \psi$) is real on $\mathbb{R}$, and $\int_{\gamma_\lambda}\psi\,d\xi = H(\xi_-) - H(\xi_+)$ is real. Hence $|D_\lambda| = \pi(1-\lambda)$. This is the half-residue of τ_λ at its pole $\xi = 0$ (residue $-(1-\lambda)$), which γ_λ passes above. For $\lambda \in \{0,1\}$, use $0 < \Phi < 1$. □

Remarks. (1) *Tools:* one trigonometric identity, $y\cot y$ decreasing, two explicit primitives (no Cauchy theorem, residue or branch cut), Euler's Beta integral. The reflection formula is not used. At $r = 1$ the integral is elementary, $\int_0^\pi K = \frac{\sin\pi a}{ab}$ because $(\sin^b ax\,\sin^a bx)' = ab\,K$, so the proof *yields* $\Gamma(a)\Gamma(1-a) = \pi/\sin\pi a$. It also gives Theorem 1 for $-1 < r < 0$. (2) *An explicit Archimedes map.* Let $\theta = x - \arg(1-e^\xi) - \operatorname{Im}H(\xi) = \operatorname{Im}Q(\xi)$, where $Q = \xi - \log(1-e^\xi) - H$ has $Q' = \frac{1}{1-e^\xi} - \psi = \frac{e^{a\xi}}{1-e^\xi}$. Differentiating $(*)$ gives $Q'(\xi)\,d\Phi\wedge d\xi = -dt\wedge d\xi$, hence $d\Phi\wedge d\theta = -dt\wedge dx$. As θ runs from 0 to π along each level arc (as in (d)), (Φ,θ) is an area-preserving bijection of R onto itself. For $a = \frac12$ it is the passage between cylindrical coordinates about two perpendicular axes of the sphere. (3) *Sources.* Conjectured by T. Amdeberhan, MathOverflow 253703 (2016); F. Petrov proved the integer case there by a contour shift and noted the equivalence with Theorem 2. In probability the identity is the moment formula of Kanter's variable (implicit in Kanter 1975, explicit in Devroye 2009; in binomial form for integer s, T. Simon 2014), proved via the Chernin–Ibragimov–Zolotarev formula; Theorem 2 then follows by the Shanbhag–Sreehari identity. The direct proof of Theorem 2 and the map in (2) are new to our knowledge. Figures, a second (finite-sum) proof and numerics: `MAIN_RESULT.pdf`.